

Complete Azure VM Creation Guide Using Python in VS Code

1. Install Azure CLI

```
az --version
az login
az account show
```

2. Create Project Folder

```
mkdir azure-vm-python
cd azure-vm-python
code .
```

3. Create Virtual Environment

```
python -m venv venv
venv\Scripts\activate
```

4. Install Packages

```
pip install azure-identity
pip install azure-mgmt-resource
pip install azure-mgmt-network
pip install azure-mgmt-compute
```

5. Get Subscription ID

```
az account show --query id --output tsv
```

6. Python Starter Code

```
from azure.identity import DefaultAzureCredential
from azure.mgmt.resource import ResourceManagementClient
from azure.mgmt.network import NetworkManagementClient
from azure.mgmt.compute import ComputeManagementClient
```

7. Create Resource Group

```
resource_client.resource_groups.create_or_update(
    RESOURCE_GROUP,
    {"location": LOCATION}
)
```

8. Create Virtual Network

```
vnet = network_client.virtual_networks.begin_create_or_update(
    RESOURCE_GROUP,
    "myVNet",
    {"location": LOCATION,
     "address_space": {"address_prefixes": ["10.0.0.0/16"]}}
).result()
```

9. Create Subnet

```
subnet = network_client.subnets.begin_create_or_update(
    RESOURCE_GROUP,
    "myVNet",
    "mySubnet",
    {"address_prefix": "10.0.0.0/24"}
).result()
```

10. Create Virtual Machine

```
vm_parameters = {
    "location": LOCATION,
    "hardware_profile": {"vm_size": "Standard_B1s"},
    "storage_profile": {
        "image_reference": {
            "publisher": "Canonical",
            "offer": "UbuntuServer",
            "sku": "18.04-LTS",
            "version": "latest"}},
    "os_profile": {
        "computer_name": VM_NAME,
        "admin_username": "azureuser",
        "admin_password": "YourPassword123!"},
}
```

11. Run Script

```
python create_vm.py
```